A novel aggregation method to promote safety security for poisoning attacks in Federated Learning

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***Abstract*—Federated Learning enables devices to collaboratively learn a shared prediction model while keeping all the training data on the local device and promoting clients' privacy. Vanilla federated learning models are susceptible to model poisoning attacks, where malicious nodes can inject fake models weight in order to deviate the global model to its objective. With this attack, malicious nodes intend to generate inaccurate global models or even to generate global models that make wrong inferences. This work proposes a new similarity function for federated learning applications to tackle the model poisoning vulnerability. Our method uses a new security aggregation proposal for local models based on the quantification of the heterogeneity of the data. In addition, this quantifier benefits from theoretical results found in models that use the auxiliary space proposed in our model. To assess the general behavior of our method, we evaluate our proposal in a federated scenario under non-iid data where all local models are honest, where we outperformed the vanilla model by 52.84% and 58.88% on two real-world dataset, respectively. In addition, our proposal reached 81.79% and 73.92% of F1-Score in a model poisoning experiment and outperformed the other state-of-the-art methods by 8.60% and 5.54%, respectively.**

***Index Terms*—Federated Learning; Deep metric learning; Similarity space; malware classification**

I. Introduction

Machine Learning (ML) has become increasingly impactful in our lives. We can see the rapid growth of ML technologies in empowering diverse applications [1]. A decentralized ML approach called Federated Learning (FL) was proposed to guarantee that training data remains on personal devices and facilitates complex models of collaborative ML on distributed devices.

In FL, edges (a.k.a. clients) use local data to train a local model. The joining of several local models estimates an ML model cooperatively in an FL server. The server receives the update of the local models, i.e., the model weights or gradients, and aggregates these models. The steps are repeated in multiple rounds until the model reaches a desirable accuracy.

Compared to conventional cloud-centric training approaches, the FL model trained at mobile edge networks features the following advantages [2]: Highly efficient use of network bandwidth, Privacy, and Low latency. However, system vulnerability can be exploited by malicious participants, which control the server model's aggregation result [3]. For example, the FL framework is vulnerable to model poisoning attacks in which some malicious end devices try to break model security by uploading fake model weights [4]. Intuitively, attacks can be defended if the aggregator checks the training process and detects anomalous local models. However, inspecting the entire training process for each end device is hard, especially when the devices are deployed on a large scale.

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On the other hand, similarity metrics are functions that provide a way to quantify the similarity between two objects. Different ML problems often require a different notion of similarity semantics, usually poorly captured by standard distance metrics (e.g., Euclidean distance).

Thus, we found several works [5, 6] using metric learning with neural networks called Deep Metric Learning (DMeL) to search for the best non-linear space that captures the semantics of the data. DMeL proves to be a promising approach to classifying data when we have no instances of particular classes in the training set [6], characteristic typically found in FL data [3]. DMeL proposals have some limitations, such as slow convergence and optimal local, which make the model's optimization a challenging task [6].

So, we proposed a new deep metric learning approach and a new auxiliary representation space, called S-space, to capture complex similarity regions such as disjoint similarity/dissimilarity regions. In this context, this paper aims to investigate the deep metric learning approach to extract relevant, robust information from heterogeneous data scenarios in FL. In this sense, *our major novelty* is proposing an interpretable quantifier for DMeL aggregation based on FL application scenarios. We used this quantifier to propose a safety learnable metric for FL applications.

We evaluate our proposal in a federated scenario without any attack, where we achieve a performance of 87.68% and 80.74% on F1-Score. These results outperformed the vanilla neural network model without our auxiliary space by 52.84% and 58.88% in a classification model on a real-world dataset. Finally, in the experiment considering model poisoning, our approach reached an F1-score of 81.79% and 73.92% for the two real datasets analyzed in the FL experiment, outperforming other methods by 8.60% and 5.54%, respectively.

This paper was organized as follows: Section II presents the related works to FL aggregation; Section III describes describes our proposal; Section IV describes the methodology used to analyze the data; Section V presents the main results;

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and Section VI concludes this work and shows some future perspectives.

II. Related work

A. Federated learning

Several works propose an FL system to estimate the importance of the contribution of each client without having access to their local data [7]. Typically, we obtained this estimate by evaluating the impact of each client on the performance of the aggregate FL model. The aggregation of local models can use clients' contribution amounts for reward distribution and fairness in FL.

Many defense methods found in the literature are based on identifying anomalous behavior arising from local models sent by customers. For the detection of anomalies, some works are based on robust aggregation functions to outliers such as median [8], mean with exclusion [9], geometric mean [10], and clustering of nearby gradients [11]. However, removing the outliers may lose valuable information from the non-malicious local models. This type of exclusion in a client with heterogeneous data can generate overfitting in the global model, thus causing poor performance in a real-world situation [3].

Analyzing federated approaches that use metric learning techniques, Yu et al. [5] proposed a generic framework for training clients that only know one class, called FedAwS. Based on the contrastive loss approach, the proposal builds a latent representation space that groups points of the same class into local groups, i.e., each client builds its own latent space. To build the aggregate model, FedAwS aggregates the local *latent spaces*, thus building a new representation that has the global context of the data. Similarly, Park et al. [12] proposes FedMetric, in which clients update local models with a loss function based on metric learning to minimize intraclass variance and simultaneously maximize interclass variance to prototype vectors generated by each of the clients.

To the best of our knowledge, we did not find any work in the literature that proposes a metric learning technique for the FL (attack) application scenario. The works discussed in this section [5, 12] use the proposed loss functions for learnable metrics for classification problems, but they do not propose any learnable metrics. Therefore, to the best of our knowledge, this work proposes the first proposal of a learnable metric in an FL scenario.

III. Our proposal

A. Data representation

Let the set $\mathcal{X} = \{\boldsymbol{x}_i\}_{i=1}^{v}$, with $\boldsymbol{x}_i \in \mathbb{R}^m$, defined in an m-dimensional feature space with v elements. For each $\boldsymbol{x}_i \in \mathcal{X}$ there is an associated label $y_i \in \mathcal{Y}$. The set $\mathcal{X}$ was called *original feature space*, and the *representation function* $f_\Theta : \mathcal{X} \to \mathcal{Z}$ obtains a new representation for the element $\boldsymbol{x}_i$ where, $f_\Theta(\boldsymbol{x}_i) = \boldsymbol{z}_i$ and $\boldsymbol{z}_i \in \mathbb{R}^d$ is a d-dimension vector. We called the set $\mathcal{Z}$ as *latent feature space*.

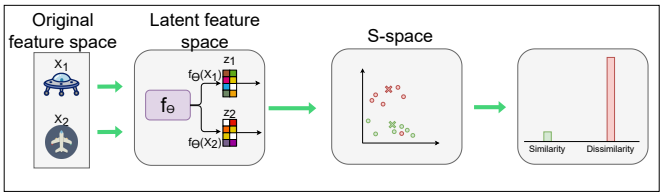


Fig. 1: Schematic of similarity extraction tailored S-space.

B. Similarity Space

We hypothesized that the *latent feature space* could be improved with an auxiliary space. Unlike the literature, we used our previous approach to quantify similarity using an auxiliary space called *S-Space* [13], as we can see in Figure 1 and we will detail it now.

We defined a function $f^S : \mathcal{X} \times \mathcal{X} \to \mathcal{S}$ that maps a pair of elements from the *original feature space* into a new auxiliary space called *similarity space* (or *S-Space*), where if $\boldsymbol{x}_i$ has the same label as $\boldsymbol{x}_j$, then $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is a similar pair. On the other hand, we consider $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ to be a dissimilar pair if $\boldsymbol{x}_i$ has a different label than $\boldsymbol{x}_j$.

We define the mapping function f^S to a pair $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ by the following pairwise operation

$$\begin{aligned} \boldsymbol{s}_{ij} &= f^S(\boldsymbol{x}_i, \boldsymbol{x}_j) = |f_\Theta(\boldsymbol{x}_i) - f_\Theta(\boldsymbol{x}_j)| \\ &= |\boldsymbol{z}_i - \boldsymbol{z}_j| = (|z_i^1 - z_j^1|, |z_i^2 - z_j^2|, \cdots, |z_i^d - z_j^d|), \end{aligned} \tag{1}$$

where $\boldsymbol{s}_{ij}$ has the same dimension as $\boldsymbol{z}_i$ and z_i^n is the n-th *feature* of the i-th sample in a *latent space representation* $\mathcal{Z}$. Finally, we use the absolute operation to preserve symmetry, i.e., $\boldsymbol{s}_{ij} = \boldsymbol{s}_{ji}$.

In *S-Space*, we have some markers to calculate the similarity between input pairs. The set of similarity markers is defined as $\mathcal{M}^+$, and similarly, markers in the set $\mathcal{M}^-$ quantify dissimilarity. So the set of all markers is $\mathcal{M} = \mathcal{M}^+ \cup \mathcal{M}^-$, where $\mathcal{M}^+ \cap \mathcal{M}^- = \emptyset$.

Therefore, for $\boldsymbol{s}_{ij}$ associated to the pair $(\boldsymbol{x}_i, \boldsymbol{x}_j)$, the closer the vector $\boldsymbol{s}_{ij}$ to the marker $\boldsymbol{\mu}^+ \in \mathcal{M}^+$, the greater the similarity value between the pairs $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$. Analogously, the distance to $\boldsymbol{\mu}^- \in \mathcal{M}^-$ measures the dissimilarity value. We can use the d function to measure the similarity between $\boldsymbol{s}_{ij}$ and the marker $\boldsymbol{\mu}_m \in \mathcal{M}$, as $q_{ij}^m = d(\boldsymbol{\mu}_m, \boldsymbol{s}_{ij})$, where q_{ij}^m is the similarity/dissimilarity value of $\boldsymbol{s}_{ij}$ and the marker μ_m. We define $q_{ij}^+ = \sum_p q_{ij}^p$ for all $\boldsymbol{\mu}_p \in \mathcal{M}^+$ and $q_{ij}^- = \sum_n q_{ij}^n$ for all $\boldsymbol{\mu}_n \in \mathcal{M}^-$. In this proposal, following [14], we use Cauchy kernel as

$$d(\boldsymbol{\mu}_m, \boldsymbol{s}_{ij}) := q_{ij}^m = \frac{\left(1 + ||\boldsymbol{s}_{ij} - \boldsymbol{\mu}_m||_2^2\right)^{-1}}{\sum_{\mu_{m'} \in \mathcal{M}} \left(1 + ||\boldsymbol{s}_{ij} - \boldsymbol{\mu}_{m'}||_2^2\right)^{-1}}$$

.

By definition, we conclude that q_{ij}^+ is the probability of the pair $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ having the same label and q_{ij}^- is the probability of the pair $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ be dissimilar (different label). Also, there are no common elements in the sets $\mathcal{M}^+$ and $\mathcal{M}^-$, i.e., we have $q_{ij}^+ + q_{ij}^- = 1$.

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Let $\mathcal{Q} = \{q_{ij}\}$, the model output, be the set that contains the pairs $q_{ij} = (q_{ij}^+, q_{ij}^-)$ corresponding to the probability of the elements of a pairwise input $(\boldsymbol{x}_i, \boldsymbol{x}_j)$ be similar or dissimilar, respectively. The loss function j can be defined as $j = \sum_{i\in\mathcal{X}} \sum_{j\in\mathcal{X}} H_c(\boldsymbol{u}_{ij}, \boldsymbol{q}_{ij}))$, where H_c is the binary cross-entropy loss and $\boldsymbol{u}_{ij} \in \mathcal{U}$ is defined as $\boldsymbol{u}_{ij} = (1, 0)$ if i has same label as j and $\boldsymbol{u}_{ij} = (0, 1)$, otherwise.

So, our proposal adds the time complexity $O(dw)$, where d is the latent feature space dimension (typically $d < 1024$, and w is the number of markers (typically $w < 8$). Furthermore, $O(dw) \ll O(N)$, where $O(N)$ is the complexity of the neural network training (encoder function).

C. Federated learning

Consider N users $\{u_1, \cdots, u_N\}$, all of whom wish to train a ML model by their respective dataset $\{\mathcal{X}_1, \cdots, \mathcal{X}_N\}$. The usual method of ML training is to group all the data in the set $\mathcal{X} = \mathcal{X}_1 \cup \cdots \cup \mathcal{X}_N$ to train a model $T_{\mathcal{X}}$. An FL system is a learning process in which the data owners collaboratively train a model T_{Fed}. In this process, any data owner u_i does not expose its data $\mathcal{X}_i$ to others. In general, the FL training process includes the following three steps[1] [2]:

- Step 1 (Task initialization): The system initializes the local models $T_{\mathcal{X}_i}$ and all necessary hyper-parameter values (e.g., learning rate) to start the learning task.
- Step 2 (Local model training and update): Each user u_i respectively uses its local data $\mathcal{X}_i$ and device to update the local model $T_{\mathcal{X}_i}$. The goal of user u_i is to find optimal parameters for the local model that minimize loss function $L(T_{\mathcal{X}_i})$, i.e., $\Theta^*_{\mathcal{X}_i} := \arg\min_{\Theta_{\mathcal{X}_i}} L(T_{\mathcal{X}_i})$, where $\Theta_{\mathcal{X}_i}$ is the parameters to model $T_{\mathcal{X}_i}$.
- Step 3 (Model aggregation and update): The server aggregates the local models from selected participants and sends the updated to aggregated model T_{Fed} and back to the data owners. Finally, we can update the server model weight values as

$$\Theta_{Fed} \leftarrow \Theta_{Fed} + \eta \sum_{i=1}^{N_{sel}} \left[p_i(\Theta^*_{\mathcal{X}_i} - \Theta_{Fed})\right], \quad (2)$$

where N_{sel} is the client's number selected in the training round, η is the learning rate, and p_i is the user-defined term. Typically, $p_i = 1/N_{sel}$ or $p_i = n_i / \sum_{i=1}^{N_{sel}} n_i$, with $n_i = |\mathcal{X}_i|$ as can see in [1].

Steps 2-3 were repeated in the training process until the aggregate loss function converged or achieved a desirable training metric.

D. Aggregation

Due to the construction of the S-space, we can obtain some theoretical proprieties.

Definition III.1. *(Optimal Latent Space) Let $\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathcal{X}$ and a latent representation function $f_\Theta : \mathcal{X} \to \mathcal{Z}$. The transformation f_Θ generates an optimal latent space $\mathcal{Z}$ when the expected value $\mathbb{E}[||\boldsymbol{s}_{ij}||_2] = 0 \implies \ell(\boldsymbol{x}_i) = \ell(\boldsymbol{x}_j)$, where function $\ell : \mathcal{X} \to \mathcal{Y}$ maps an unlabeled example $\boldsymbol{x}_i$ into their respective label y_i.*

Our proposal can group points of the same class into clusters. It is worth noting that we defined the optimal space as a conditional instead of a biconditional statement. From this definition, we can observe that our model may create several different clusters of the same class.

Proposition III.1. *In S-space, given k positive markers in the set $\mathcal{M}^+$, $n-k$ negative markers in $\mathcal{M}^-$ and $k = n - k$, the latent space found by our approach, i.e., the estimation of the parameters Θ of f_Θ, generates an optimal latent space if $\exists\ \boldsymbol{\mu}_i \in \mathcal{M}^+$ so that $||\mu_i||_2^2 < ||\mu_j||_2^2$ for any $\boldsymbol{\mu}_j \in \mathcal{M}^-$.*

Proof. Given $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, our proposal measures the similarity between the entries through the Cauchy kernel given by q_{ij}^+, so that for $\boldsymbol{\Sigma}^*$ it follows that $q_{ij}^+ = 1 \iff \ell(x_i) = \ell(x_j)$.

Since f_Θ generates an optimal space, we then have $\mathbb{E}[||s_{ij}||_2] = 0 \implies \ell(x_i) = \ell(x_j)$, so, it follows that for a optimal latent space, we must have

$$\begin{aligned} q_{ij}^+ &= \frac{\sum_{k\in\mathcal{M}^+}(1 + ||\mu_k||_2^2)^{-1}}{\sum_{k\in\mathcal{M}^+}(1 + ||\mu_k||_2^2)^{-1} + \sum_{s\in\mathcal{M}^-}(1 + ||\mu_s||_2^2)^{-1}} \\ &= \frac{1}{1 + \frac{\sum_{s\in\mathcal{M}^-}(1+||\mu_s||_2^2)^{-1}}{\sum_{k\in\mathcal{M}^+}(1+||\mu_k||_2^2)^{-1}}}. \end{aligned}$$

Hence, if we want $\ell(x_i) = \ell(x_j)$, we should ideally have q_{ij}^+ tends to 1. It follows that $\sum_{s\in\mathcal{M}^-}(1 + ||\mu_s||_2^2)^{-1} < \sum_{k\in\mathcal{M}^+}(1 + ||\mu_k||_2^2)^{-1}$. Therefore, let μ^+ be the element with the smallest module in the set $\mathcal{M}^+$; we then have $\sum_{k\in\mathcal{M}^+}(1 + ||\mu_k||_2^2)^{-1} < k(1 + ||\mu^+||_2^2)^{-1}$. Analogously, we can consider μ^- as the vector with the largest module in the set $\mathcal{M}^-$, so, $\sum_{k\in\mathcal{M}^-}(1+||\mu_k||_2^2)^{-1} > (n-k)(1+||\mu^-||_2^2)^{-1}$.

We can then conclude that $k(1 + ||\mu^+||_2^2)^{-1} > (n-k)(1 + ||\mu^-||_2^2)^{-1}$, and therefore, $(n-k)(1+||\mu^+||_2^2) < k(1+||\mu^-||_2^2)$. Furthermore, adding the restriction that our propose has a same count of positive and negative markers ($k = n - k$), we have $1 + ||\mu^+||_2^2 < 1 + ||\mu^-||_2^2$. □

Thus, by the **Proposition** III.1, given the positive marker with the smallest module $||\boldsymbol{\mu}^+||_2$, we get information about the separability of the *latent feature space*. So, in this paper, we propose a novel function to quantify the separability of *latent feature space* (called model inference ability) as

Definition III.2. *We define the inference ability K of a model based on S-space $\mathcal{S}$ as being*

$$K(\mathcal{S}) = 1 - \frac{||\boldsymbol{\mu}^+||_2}{||\boldsymbol{\mu}^-||_2},$$

where $||\boldsymbol{\mu}^+||_2 \leq ||\boldsymbol{\mu}||_2$ for all $\boldsymbol{\mu} \in \mathcal{M}^+$ and $||\boldsymbol{\mu}^-||_2 \leq ||\boldsymbol{\mu}||_2$ for all $\boldsymbol{\mu} \in \mathcal{M}^-$.

Therefore, let $\mathcal{S}_i$ be the similarity space found by the FL user model $T_{\mathcal{X}_i}$ associated with the user u_i, we can

[1]the local model refers to the model trained at each user u_i, whereas the global (a.k.a. federated) model refers to the model aggregated T_{Fed}

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calculate the inference capacity as being $K(\mathcal{S}_i)$. Given an FL application, we can build the aggregate model described in Equation 2 as $\Theta_{Fed} \leftarrow \Theta_{Fed} + \eta \sum_{i=1}^{N_{sel}} \left[j_i(\Theta^*_{\mathcal{X}_i} - \Theta_{Fed}) \right]$, where $j_i = K(\mathcal{S}_i) / \sum_{s=1}^{N_{sel}} K(\mathcal{S}_s)$.

Additionally, we assume a network classification scenario where all clients trains the same network model, but using their own data. It is not uncommon to see that network characteristics differ across different networks. That said, the resulting models will be abstractions of different real-world conditions. So, our proposed solution handles such situations, and such differences are not considered malicious because our proposal quantifies data adherence to the model.

IV. Methodology

A. Model attack

A Θ^*_{Mal} model is said to be malicious when it deliberately fulfills its intention to harm a machine learning system. Bagdasaryan et al. [15] defines a malicious model that aims to replace global model Θ_{Fed} to introduce error (e.g., bad functionality) in the new model Θ_{Poi}. Thus, given a FL round with $\{T_i\}_{i=1}^{N_{sel}-1}$ local non-malicious model and a malicious model T_{Mal}, the aggregation (Equation 2) can be defined in this attack as

$$\Theta_{Poi} \leftarrow \Theta_{Fed} + \eta \left[\sum_{i=1}^{N_{sel}-1} \left[p_i(\Theta^*_{\mathcal{X}_i} - \Theta_{Fed}) \right] + p_{Mal}(\Theta^*_{Mal} - \Theta_{Fed}) \right],$$

where Θ_{Poi} is the new global federation model.

Thus following [3, 15], when Θ_{Fed} converges, we have that $\sum_{i=1}^{N_{sel}-1} \left[p_i(\Theta^*_{\mathcal{X}_i} - \Theta_{Fed}) \right] \approx 0$. Thus, a malicious client model Θ^*_{Mal} can replace the global model Θ_{Fed} with a poisoning model Θ_{Poi} as

$$\Theta^*_{Mal} = \Theta_{Fed} + \frac{1}{\eta p_{Mal}} (\Theta_{Poi} - \Theta_{Fed}).$$

In this work, we use $\eta = 0.01$, as can see in [15]. In addition, we defined ϵ as the proportion of malicious clients in the federation.

B. Dataset

To analyze the federated performance of our proposal, we used two datasets:

- **Malimg Dataset** was developed by the University of California's vision research laboratory[2]. This dataset contains 9339 samples from 25 malware families, obtained through experiments of mixtures of network and the Windows operating system malware. The Malimg dataset has diversity in the number of examples. Specifically, the number of samples from the *malware* families ranges from 80 to 2949.
- **MaleVis Dataset** is a corpus involving byte images of 26 (25+1) classes. Here, 1 class represents the goodware samples while the rest of the 25 classes correspond to different malware types. MaleVis dataset involves a total of 14226 images. For the MaleViz dataset, malware examples range from 470 to 500. The Maleviz dataset contains multiple classes with 25 malware families and 1832 goodware examples.

Each binary code of *malware* is a vector of 8-bit unsigned integers, which is organized in a two-dimensional matrix resized to 64 x 64 and visualized as a gray-scale image.

C. Model evaluation

We use the framework Flower [16] to develop solutions and applications in the context of FL. We separate the dataset into training (users), validation (server), and test (server), with proportions of 80%, 10%, and 10%, respectively. We performed a non-uniform identically distributed among the users (training) used in this experiment, i.e., we randomly distributed the *malware* code samples in a non-uniform way (quantity-based label imbalance), as can see in McMahan et al. [17]. Thus, we used a server and 40 clients to evaluate our model, and we trained our method with an NVIDIA Quadro RTX 6000 GPU (24 GB) for a total of 125 epochs (server). For each training round, the server selects five clients to train the local model, i.e., each model is trained using only the local data. Finally, the models are aggregated on the central server, which forwards the aggregated model to the clients.

We use the K-nearest neighbors (KNN) classifier to evaluate our approach, with three neighbors, according to [6]. The performance of the KNN classification can sometimes be significantly improved through supervised similarity functions. We calculated the experiment's accuracy and F1-Score.

D. Network architecture

We initialize the position of markers with Lloyd's algorithm, and for the encoder, according to [13], we set network dimensions to m-500-500-2000-d for all datasets, where m is the number of features of the input data, and d is the *latent space representation* dimension. For this work, we use d = 64. All layers of neural networks are fully connected, and we use ReLU as the activation function. In addition, we use Stochastic Descending Gradient (SGD) with momentum, with a learning rate of 0.01 and a momentum of 0.9.

The optimization model depends on some hyper-parameters (n, k), so we investigated which value of these variables could maximize the model's accuracy. We used the grid search technique for hyper-parameter optimization and we found $k = 3$ similarity markers and $n - k = 3$ dissimilarity markers. These values were used during the rest of the work.

V. Results and Discussion

A. Qualitative results

Initially, for Malimg dataset, we evaluated the performance of the K quantifier proposed in this work. We evaluated this quantifier during the client training, as shown in Figure 2.

This experiment shows a negative correlation between the value of *loss* and the model inference ability K. *Loss* function quantifies a measure of dissimilarity between the model result and the label, i.e., the lower the value of *loss*, the better the model performance for the analyzed data. However, *loss*

[2]https://vision.ece.ucsb.edu/research/signal-processingmalware-analysis

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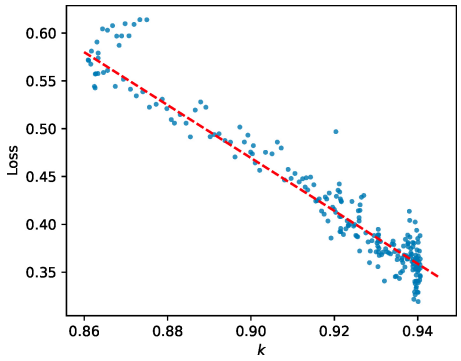


Fig. 2: Loss versus inference ability K.

is dependent on the dataset, so if a change is made in the evaluation dataset, the value of *loss* is modified. Note that this behavior is not observed in our proposal, as the quantifier is defined to the model markers, so this quantifier is agnostic to the validation dataset used by our proposal.

B. Quantitative result

In Table I and II, we performed an ablation study in our proposal and, in this evaluation, we compare our method with three approaches: (i) We use a NN with a federated average aggregation approach; (ii) We use a NN and markers approach with federated aggregation average approach and (iii) we use our proposal (section III-D). The server never had access to user data, only local models. We use the models trained for 100 epochs, and malicious clients attack the models in the remaining 25 rounds.

Initially, for none malicious client ($\epsilon = 0$), we have that NN without markers (i) has the worst performance for both datasets among all the analyzed approaches, with an F1 measure of 29.01% and 20.22%, surpassing by the two versions in ablation experiment. The NN and marker (ii) approach using FedAvg aggregation has the second-best performance for F1-Score with 84.79% for the Malimg dataset (Table I) and 79.92% for the MaleViz dataset (Table II). Finally, our aggregation proposal (iii) achieved the best F1-Score performance for Malimg and MaleViz datasets with 87.68% and 80.74%.

However, when we added the malicious clients, we noticed that the FedAvg aggregation algorithm had a degraded result. For $\epsilon = 0.125$ in the MaleViz dataset, we see that NN w/ marker approach (ii) get a value of 52.23% and 51.72% for ACC and F1-Score metrics respectively. Considering the same ϵ value, our proposal gets 79.95% and 79.46% for the same metrics. Finally, for $\epsilon = 0.25$, we noticed that NN w/ marker (ii) approach obtains a 43.60% for the ACC metric, resulting in a difference equal to 31.18% when compared to our proposal (iii). This difference is more evident when we compare the F1-Score metric, where we observe a difference between the two approaches of 35.21%. A similar conclusion is observed in Malimg dataset in Table I.

In addition, we evaluated the performance of our proposal versus different model aggregation approaches found in the literature. We observed that our proposal performs best for the two metrics analyzed. For $\epsilon = 0.125$, NN w/ markers + Krum-mean aggregation obtained the second-best performance for Malimg dataset, with ACC and F1-Score equal to 90.26% and 76.83%, respectively in Malimg dataset. Our proposal achieves the best results compared to all other techniques, reaching 93.15% and 87.68%.

For $\epsilon = 0.25$, the second and third best ACC results were obtained by NN w/ markers + Median and NN w/ markers + Trimmed-mean with 80.24% and 79.98%, a difference of 9.29% and 9.55% to our proposal, respectively. Similarly, for the F1-score metric, our approach obtained 81.79%, the best overall result. The second and third best approaches obtained 73.19% and 67.49%, by NN w/ markers + FedEqual, and NN w/ markers + Krum-mean, respectively. A similar conclusion is observed in the MaleViz dataset in Table II.

Finally, we compared our approach with the centralized proposal, i.e., we merged all local client data to train a single model and evaluated this model with the data server. This centralized approach performed very similarly to our proposal but did not consider users' privacy. Thus, our method effectively performs a classification task in a federated scenario, preserving clients' data privacy.

VI. Conclusion and Future Work

We propose a new similarity function based on deep metric learning for FL scenarios in this work. Our proposal defines a quantifier to perform a security model aggregation tailored to S-space. Therefore, through the markers' position in the

Method	$\epsilon = 0$		$\epsilon = 0.125$		$\epsilon = 0.25$	
	ACC	F1-Score	ACC	F1-Score	ACC	F1-Score
(i) NN w/o markers + FedAvg	40.09%	29.01%	–	–	–	–
NN w/ markers + Trimmed-mean [9]	89.83%	76.31%	80.99%	67.63%	79.98%	61.86%
NN w/ markers + FedEqual [3]	90.72%	83.27%	82.53%	74.26%	76.62%	73.19%
NN w/ markers + Median [8]	91.43%	83.61%	81.19%	71.51%	80.24%	65.38%
NN w/ markers + Krum-mean [11]	92.61%	82.39%	90.26%	76.83%	72.04%	67.49%
(ii) NN w/ markers + FedAvg	92.93%	84.79%	65.68%	57.99%	50.66%	42.07%
(iii) Our proposal	**93.15%**	**87.68%**	**92.56%**	**83.12%**	**89.53%**	**81.79%**
NN w/ marker (Centralized)	96.17%	90.94%	–	–	–	–

TABLE I: Results of Malimg dataset and best values are in bold.

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Method	$\epsilon = 0$		$\epsilon = 0.125$		$\epsilon = 0.25$	
	ACC	F1-Score	ACC	F1-Score	ACC	F1-Score
(i) NN w/o markers + FedAvg	21.62%	20.22%	–	–	–	–
NN w/ markers + Trimmed-mean [9]	79.71%	79.30%	70.65%	71.92%	67.06%	64.91%
NN w/ markers + FedEqual [3]	78.97%	77.90%	74.46%	73.42%	68.27%	68.38%
NN w/ markers + Median [8]	78.25%	76.94%	72.32%	72.06%	63.84%	62.21%
NN w/ markers + Krum-mean [11]	79.13%	78.45%	73.63%	73.24%	67.89%	67.75%
(ii) NN w/ markers + FedAvg	**80.50%**	79.92%	52.23%	51.72%	43.60%	38.71%
(iii) Our proposal	80.09%	**80.74%**	**79.95%**	**79.46%**	**74.78%**	**73.92%**
NN w/ marker (Centralized)	83.31%	82.79%	–	–	–	–

TABLE II: Results of MaleViz dataset and best values are represented in bold.

similarity space, we obtained an interpretable quantifier about the heterogeneity of the data. Furthermore, our proposal is agnostic to the validation dataset. We have conducted various experiments over the analyzed dataset and showed that our approach consistently outperforms other methods. Our model showed evidence of robustness to attacks, showing a gain of 35.21% against the standard neural network model and a difference of 8.60% for the F1-Score compared to aggregation proposals found in the literature. In future work, we intend to investigate the self-supervised learning scenario because these approaches tend to act in neural networks as a regularizer, thus increasing the generalization of the proposal, making it perform well in zero-shot learning tasks and non-identically distributed problems, typically found in FL applications.

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